

Synthesis of 1,4- and 1,5-Amino Alcohols via Nucleophilic Addition of Semicyclic *N,O*-Acetal with Organozinc Reagents

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Supporting Information

ABSTRACT: An efficient approach to access functionalized 1,4- and 1,5-amino alcohols has been developed through the nucleophilic addition of semicyclic *N,O*-acetal with organozinc reagents. A number of substituted benzyl zinc reagents (including nitrile and ester substituted) could react with semicyclic *N,O*-acetals **1** and **2**, affording the desired products **3a–3p** and **4a–4o** in good to excellent yields.



The development of an efficient methodology to access privileged structural motifs is one of the most important tasks in contemporary organic chemistry.¹ As a prime instance, 1,4- or 1,5-amino alcohols exist as subunits in many natural products and many biologically active molecules in the life-science industry.² In particular, functionalized 1,4- or 1,5-amino alcohols are used as important materials and serve as flexible building blocks in synthetic chemistry.³ Notably, 1,4-amino alcohols are also applied as versatile ligands for numerous metals in the field of catalysis.⁴ Recently, there has been great progress in the synthesis of 1,2-amino alcohols,⁵ whereas there are only a few new synthetic approaches to 1,4- or 1,5-amino alcohols, including reductive *N*–*O* bond cleavage of 1,2-oxazines,⁶ addition of a Grignard reagent to the 2-tosylaminotetrahydrofuran process,⁷ Lewis-acid-catalyzed ring-opening reaction of semicyclic *N,O*-acetal with silyl enol ethers^{2b} or the 1,3-dicarbonyl derivative,^{8a} asymmetric Mannich reactions of α -thio acetaldehydes,^{8b} and addition of amines to 2-sulfinyl dienes.⁹ Owing to the high reactivity of Grignard reagent or difficulty in preparing the reaction substrate, the practical methods to functionalized 1,4- and 1,5-amino alcohols are very limited.

In the past decade, functionalized organozinc reagents, which were easily prepared through an insertion reaction of zinc metal into various substituted organic halides, have demonstrated many advantages in modern organic synthesis.¹⁰ Many useful transformations from functionalized organozinc reagents were successfully realized through the addition to carbonyl¹¹ and nitrones,¹² as well as addition–elimination¹³ or addition–migration.¹⁴ More recently, a coupling reaction of organozinc reagents with *N,O*-acetals was reported.¹⁵ On the basis of our efforts in the transformation of *N,O*-acetals,¹⁶ we envisioned that semicyclic *N,O*-acetals could undergo nucleophilic addition from functionalized organozinc reagents to give 1,4- and 1,5-amino alcohols (Figure 1). As a continuation of our interest in developing a novel method for useful intermediates,¹⁷ herein we present a practical method to functionalized 1,4- and 1,5-amino alcohols through a nucleophilic addition process from various substituted organozinc reagents with semicyclic *N,O*-acetals **1** and **2**.

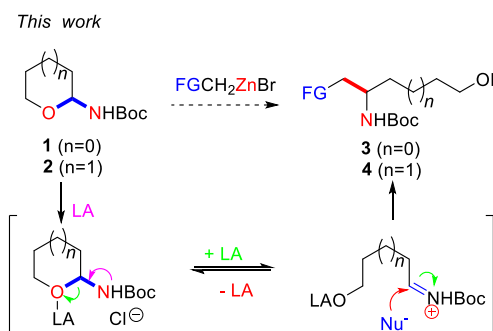


Figure 1. Our new strategy to access 1,4- and 1,5-amino alcohols.

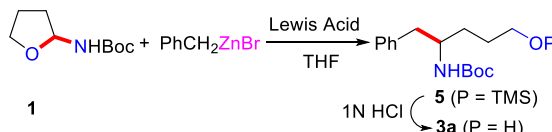
Our investigation started with the reaction of semicyclic *N,O*-acetal **1** with benzylzinc bromide. When the mixture of **1** and benzylzinc bromide was treated at $-78\text{ }^{\circ}\text{C}$ to room temperature overnight, no desired product **3a** was observed (Table 1, entry 1). When the mixture was treated with Lewis acid $\text{BF}_3\cdot\text{OEt}_2$ at $-78\text{ }^{\circ}\text{C}$ for 12 h, **3a** was produced in a low yield (Table 1, entry 2). When TMSOTf was investigated, **3a** was produced in a 76% yield (Table 1, entry 3). While the reaction was conducted at $0\text{ }^{\circ}\text{C}$, the yield was significantly reduced (Table 1, entry 4). TiCl_4 led to a similar yield of **3a** (Table 1, entry 5). To improve the reaction yield, various Lewis acids were screened, and the results were summarized in Table 1 (Table 1, entries 6–10). Delightfully, TMSCl significantly increased the yield of **3a** up to 89% (Table 1, entry 10). However, the use of 1 equiv TMSCl resulted in a significant drop of the yield of **3a** (Table 1, entry 11).

Next, we turned to investigate the scope and limitation of the nucleophilic addition of organozinc reagents and semicyclic *N,O*-acetal **1** (Scheme 1). Different substituted (methyl, fluoro, trifluoromethyl and chloro) benzyl zinc bromides were surveyed under the optimal conditions, as summarized in Scheme 1. In general, all of these *ortho*-, *meta*- and *para*-

Received: June 10, 2019

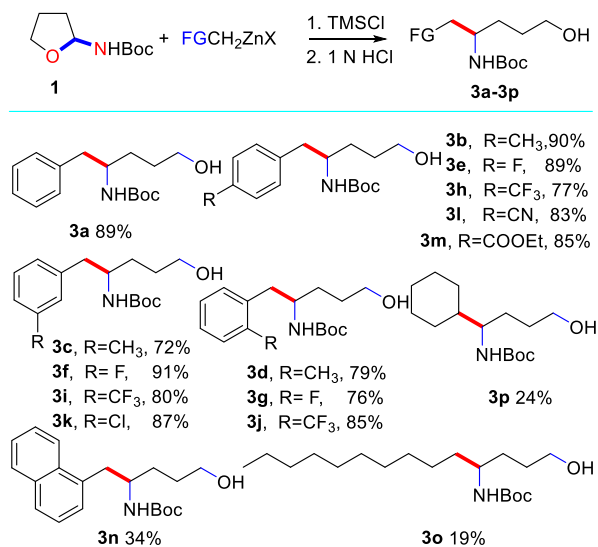
Published: August 12, 2019

Table 1. Optimization of Reaction Conditions



entries ^a	LA (equiv)	temp (°C)	time (h)	product	yield (%) ^b
1		−78 to rt	12	3a	NR
2	BF ₃ ·OEt ₂ (2)	−78	12	3a	trace
3	TMSOTf (2)	−78	12	3a	76
4	TMSOTf (2)	0	2	3a	44
5	TiCl ₄ (2)	−78	12	3a	67
6	Cu(OTf) ₂ (2)	−78 to rt	12	3a	NR
7	Sc(OTf) ₃ (2)	−78 to rt	12	3a	NR
8	In(OTf) ₃ (2)	−78 to rt	12	3a	NR
9	ZnCl ₂ (2)	−78 to rt	12	3a	NR
10	TMSCl (2)	−78	12	3a	89
11	TMSCl (1)	−78	12	3a	37
12 ^c	TMSCl (2)	−78	12	3a	86

^aThe reactions were performed with **1** (0.53 mmol) in THF (3 mL), PhCH₂ZnBr (4 equiv), and LA at −78 °C to rt for 2–12 h. The reaction was quenched with 1 N HCl solution (0.6 mL) at 0 °C for 15 min. ^bIsolated yield. ^cLarger scale synthesis of compound **3a** (5.8 g) starting with compound **1** (4.6 g).

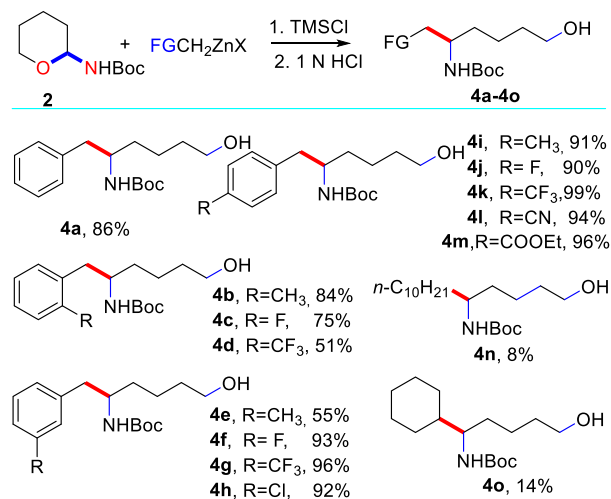
Scheme 1. Syntheses of Functionalized 1,4-Amino Alcohols^{a,b}

^aThe reactions were performed with **1** (0.53 mmol) in THF (3 mL), FGCH₂ZnBr or FGCH₂ZnI (4 equiv), and TMSCl (2 equiv) at −78 °C for 12 h. The reaction was quenched with 1 N HCl solution (0.6 mL) at 0 °C for 15 min. ^bIsolated yield.

substituted benzylzinc reagents could smoothly react with semicyclic *N,O*-acetal **1**, affording the desired 1,4-amino alcohols **3b–3p** in moderate to excellent yields. It is worth mentioning that nitrile and ester substituted benzylzinc bromides could successfully provide the products **3l** and **3m** in good yields. When 1-naphthalene zinc bromide was used, the desired product **3n** was produced in 34% yield. *n*-Decylzinc bromide and cyclohexylzinc iodide were also screened, both affording the desired products **3o** and **3p**, albeit in low yields.

Then, the scope and limitation of the nucleophilic addition for semicyclic *N,O*-acetal **2** was also explored, and the results

are summarized in Scheme 2. Most substituted benzylzinc bromides, including nitrile and ester substituted ones, could

Scheme 2. Syntheses of Functionalized 1,5-Amino Alcohols^{a,b}

^aThe reactions were performed with **2** (0.50 mmol) in THF (3 mL), FGCH₂ZnBr or FGCH₂ZnI (4 equiv), and TMSCl (2 equiv) at −78 °C for 12 h. The reaction was quenched with 1 N HCl solution (0.6 mL) at 0 °C for 15 min. ^bIsolated yield.

afford the desired products **4a–4m** in moderate to excellent yields. Other zinc reagents such as cyclohexylzinc iodide also led to the nucleophilic addition product, but in a low yield (**4o**).

According to the previous ring-opening process of semicyclic *N,O*-acetals with the silyl enol ether,^{2b} TMSCl could activate alkylzinc halide^{10b} to facilitate this nucleophilic addition process. A possible mechanism was illustrated in Figure 2.

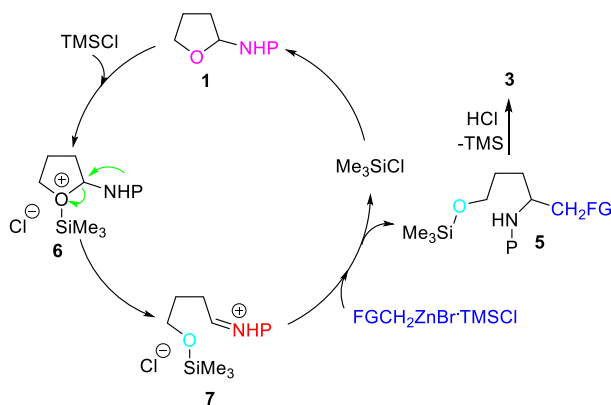


Figure 2. Proposed mechanism of the nucleophilic addition (P = Boc).

First, TMSCl (1 equiv) attached to the oxygen of the semicyclic *N,O*-acetal **1** to form a stable ring-opening *O*-TMS ether **6**, which could undergo rearrangement to form acyclic iminium intermediate **7**. Another TMSCl (1 equiv) was consumed to activate zinc reagents.^{10b} The activated nucleophilic benzylzinc then attacked the iminium intermediate **7** to afford the addition product **5** and partial TMSCl, which was continuous participation in the cyclic reaction.

In summary, we have established a new and practical approach for the synthesis of functionalized 1,4- and 1,5-amino alcohols **3a–3p** and **4a–4o** by nucleophilic addition of semicyclic *N,O*-acetals **1** and **2** with organozinc reagents. TMSCl was found to be the most effective reagent for this addition process, and a variety of substituted benzyl-containing 1,4- and 1,5-amino alcohols were successfully synthesized in moderate to excellent yields. To the best of our knowledge, the present process is the first direct method for the preparation of 1,4- and 1,5-amino alcohols containing sensitive functional groups. Moreover, this is the first example that TMSCl could catalyze the addition or substitution reaction of semicyclic *N,O*-acetals.

EXPERIMENTAL SECTION

General Considerations. THF was distilled from sodium/benzophenone, and DCM was distilled from phosphorus pentoxide. Zinc dust was activated by being stirred for 15 min with HCl (2 N), followed by washing successively with water, ethyl alcohol, acetone, and ether. Reactions were monitored by TLC-FID. Flash chromatography was performed on silica gel (300–400 mesh) with petroleum/EtOAc as the eluent. HRMS was conducted on a Thermo Scientific LTQ Orbitrap XL apparatus. IR spectra were measured using a film on a Fourier transform infrared spectrometer. NMR spectra were recorded at 400 MHz, and chemical shifts are reported in δ (ppm) referenced to an internal TMS standard for ^1H NMR and CDCl_3 (77.16 ppm) for ^{13}C NMR. The heat source was an oil bath.

Preparation of Benzylzinc Bromide. Activated zinc powder (393 mg, 6 mmol) was suspended in anhydrous THF (3.5 mL) under an argon atmosphere, and then 1,2-dibromoethane (25 μL , 5% mmol) was added dropwise. The turbidity mixture was kept at 70 $^\circ\text{C}$ for 10 min to produce vigorous bubbles. Then, TMSCl (10 μL) was added with a large number of bubbles; the supernatant of the system was quickly clear and transparent, and the bottom zinc powder was kept loose and porous. After the mixture was cooled to 0 $^\circ\text{C}$, a solution of bromide (5 mmol) in anhydrous THF (1.5 mL) was added dropwise. The resulting mixture was stirred for 10 h at 0 $^\circ\text{C}$ (as long as the reaction of zinc powder is almost complete) to give the benzylzinc bromide as 1 M/THF.

Preparation of Alkylzinc Iodides. Zinc powder (393 mg, 6 mmol) was suspended in anhydrous THF (3.5 mL) under an argon atmosphere, and then 1,2-dibromoethane (25 μL , 5% mmol) was added. The mixture was heated at 70 $^\circ\text{C}$ for 10 min. After the mixture was cooled to rt, TMSCl (25 μL , 5% mmol) was added, and the mixture was stirred for another 15 min before the solution of iodide (10 mmol) in anhydrous THF (1.5 mL) was added dropwise. The resulting mixture was stirred for 12 h at 50 $^\circ\text{C}$ to give the alkylzinc iodide as 1 M/THF.

General Procedure for the Preparation of Semicyclic *N,O*-Acetals **1 and **2**.** To a solution of a *tert*-butyl carbamate (5.00 g, 42.7 mmol) and 2,3-dihydrofuran or 3,4-dihydro-2*H*-pyran (1.2 equiv) in dichloromethane (1.0 M) was added *p*-toluenesulfonic acid monohydrate (1 mol %) at rt. The reaction mixture was stirred for 3 h. The mixture was quenched with saturated aqueous NaHCO_3 and extracted with ethyl acetate. The organic layer was washed with brine, dried over anhydrous anhydrous MgSO_4 , and concentrated *in vacuo*. The residue was purified by chromatography on silica gel (PE/EA = 6:1) to give semicyclic *N,O*-acetals **1** or **2**.

***tert*-Butyl (Tetrahydrofuran-2-yl)carbamate (**1**):**^{2b} white solid (4.54 g, 57%), mp 114–116 $^\circ\text{C}$; IR (film) ν_{max} 3330, 2975, 1685, 1514, 1363, 1161, 1040 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3) δ 5.57–5.47 (m, 1H), 5.04 (brs, 1H), 3.94–3.87 (m, 1H), 3.84–3.77 (m, 1H), 2.22–2.13 (m, 1H), 1.97–1.88 (m, 2H), 1.69–1.60 (m, 1H), 1.46 (s, 9H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.0, 82.7, 80.1, 67.0, 31.9, 28.5, 24.8 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_9\text{H}_{17}\text{NO}_3\text{Na}^+$ [$\text{M} + \text{Na}$] $^+$ 210.1101, found 210.1105.

***tert*-Butyl (Tetrahydro-2*H*-pyran-2-yl)carbamate (**2**):**^{2b} white solid (7.74 g, 90%), mp 106–108 $^\circ\text{C}$; IR (film) ν_{max} 3301, 2933,

2855, 1714, 1528, 1363, 1246, 1163, 1079, 1031, 930, 895, 878 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3) δ 5.13 (brs, 1H), 4.85–4.73 (m, 1H), 4.01–3.91 (m, 1H), 3.60–3.46 (m, 1H), 1.86–1.73 (m, 2H), 1.63–1.46 (m, 3H), 1.43 (s, 9H), 1.37–1.26 (m, 1H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 154.8, 80.1, 79.7, 67.2, 31.6, 28.4, 25.2, 23.1 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{10}\text{H}_{19}\text{NO}_3\text{Na}^+$ [$\text{M} + \text{Na}$] $^+$ 224.1257, found 224.1252.

General Procedure for the Synthesis of **3a–3p** and **4a–4o**.

To a solution of semicyclic *N,O*-acetal **1** (100 mg, 0.53 mmol) or **2** (100 mg, 0.50 mmol) in anhydrous THF (3 mL) was added dropwise freshly prepared substituted benzylzinc bromide reagents or alkylzinc iodides (4eq, 1 M in THF) at -78 $^\circ\text{C}$ under an argon atmosphere. Then TMSCl (2eq) was added, and the mixture was stirred for 12 h. The reaction was quenched with 1 N HCl solution (0.6 mL) at 0 $^\circ\text{C}$ and diluted with EtOAc (5 mL) and water (5 mL). The organic layer was separated, and the aqueous phase was extracted with EtOAc three times. The combined organic layers were washed with brine, dried over anhydrous MgSO_4 , filtered, and concentrated under reduced pressure. The residue was purified by chromatography on silica gel (PE/EA = 2:1) to give the compounds **3a–3p** and **4a–4o**.

***tert*-Butyl (5-Hydroxy-1-phenylpentan-2-yl)carbamate (**3a**):** white solid (132 mg, 89%), mp 68–70 $^\circ\text{C}$; IR (film) ν_{max} 3336, 2927, 1685, 1454, 1366, 1246, 1172, 1058, 695 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.31–7.16 (m, 5H), 4.47–4.21 (m, 1H), 3.95–3.80 (m, 1H), 3.66–3.60 (m, 2H), 2.83–2.72 (m, 2H), 1.83 (s, 1H), 1.66–1.55 (m, 3H), 1.43–1.36 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.8, 138.3, 129.6, 128.5, 126.5, 79.3, 62.7, 51.5, 41.7, 31.0, 29.2, 28.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{16}\text{H}_{26}\text{NO}_3^+$ [$\text{M} + \text{H}$] $^+$ 280.1907, found 280.1909.

***tert*-Butyl (5-Hydroxy-1-(*p*-tolyl)pentan-2-yl)carbamate (**3b**):** white solid (140 mg, 90%), mp 65–67 $^\circ\text{C}$; IR (film) ν_{max} 3336, 2979, 2927, 1687, 1514, 1363, 1248, 1174, 1056 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.12–7.04 (m, 4H), 4.48–4.28 (m, 1H), 3.89–3.75 (m, 1H), 3.64–3.59 (m, 2H), 2.79–2.66 (m, 2H), 2.31 (s, 3H), 1.98 (s, 1H), 1.67–1.54 (m, 3H), 1.45–1.36 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.9, 135.9, 135.0, 129.5, 129.1, 79.3, 62.6, 51.4, 41.1, 30.8, 29.2, 28.5, 21.1 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{17}\text{H}_{28}\text{NO}_3^+$ [$\text{M} + \text{H}$] $^+$ 294.2064, found 294.2067.

***tert*-Butyl (5-Hydroxy-1-(*m*-tolyl)pentan-2-yl)carbamate (**3c**):** white solid (112 mg, 72%), mp 82–84 $^\circ\text{C}$; IR (film) ν_{max} 3323, 2919, 1687, 1526, 1363, 1170 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.22–7.14 (m, 1H), 7.10–6.93 (m, 3H), 4.49–4.19 (m, 1H), 3.92–3.75 (m, 1H), 3.68–3.60 (m, 2H), 2.83–2.65 (m, 2H), 2.33 (s, 3H), 1.79 (s, 1H), 1.67–1.55 (m, 3H), 1.45–1.36 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.8, 138.2, 138.0, 130.4, 128.4, 127.2, 126.6, 79.3, 62.7, 51.5, 41.6, 31.0, 29.2, 28.5, 21.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{17}\text{H}_{28}\text{NO}_3^+$ [$\text{M} + \text{H}$] $^+$ 294.2064, found 294.2068.

***tert*-Butyl (5-Hydroxy-1-(*o*-tolyl)pentan-2-yl)carbamate (**3d**):** white solid (123 mg, 79%), mp 76–78 $^\circ\text{C}$; IR (film) ν_{max} 3327, 2933, 2863, 1689, 1390, 1366, 1250, 1174, 1052, 740 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.17–7.09 (m, 4H), 4.55–4.34 (m, 1H), 3.91–3.69 (m, 1H), 3.66–3.59 (m, 2H), 2.81–2.70 (m, 2H), 2.34 (s, 3H), 1.98 (s, 1H), 1.69–1.54 (m, 3H), 1.46–1.30 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.8, 136.7, 130.5, 130.3, 126.6, 125.9, 79.3, 62.6, 50.7, 39.5, 31.4, 29.2, 28.5, 19.7 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{17}\text{H}_{28}\text{NO}_3^+$ [$\text{M} + \text{H}$] $^+$ 294.2064, found 294.2069.

***tert*-Butyl (1-(4-Fluorophenyl)-5-hydroxypentan-2-yl)carbamate (**3e**):** white solid (140 mg, 89%), mp 73–75 $^\circ\text{C}$; IR (film) ν_{max} 3344, 2931, 1685, 1508, 1368, 1223, 1170, 1052 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.18–7.10 (m, 2H), 7.03–6.92 (m, 2H), 4.46–4.26 (m, 1H), 3.89–3.75 (m, 1H), 3.66–3.60 (m, 2H), 2.79–2.69 (m, 2H), 1.90 (s, 1H), 1.67–1.54 (m, 3H), 1.43–1.37 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 161.7 (d, $J = 244.1$ Hz), 155.8, 133.9, 130.9 (d, $J = 7.8$ Hz), 115.2 (d, $J = 21.2$ Hz), 79.4, 62.6, 51.6, 41.0, 31.0, 29.1, 28.5 ppm; ^{19}F NMR (376 MHz, CDCl_3) δ –117.0 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{16}\text{H}_{25}\text{FNO}_3^+$ [$\text{M} + \text{H}$] $^+$ 298.1813, found 298.1815.

tert-Butyl (1-(3-Fluorophenyl)-5-hydroxypentan-2-yl)carbamate (3f): white solid (144 mg, 91%), mp 80–82 °C; IR (film) $\nu_{\max}$ 3354, 2919, 1683, 1520, 1442, 1368, 1250, 1174, 1044, 782 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3) δ 7.28–7.21 (m, 1H), 6.99–6.86 (m, 3H), 4.45 (d, J = 8.4 Hz, 1H), 3.90–3.78 (m, 1H), 3.66–3.61 (m, 2H), 2.81–2.71 (m, 2H), 1.98 (s, 1H), 1.67–1.54 (m, 3H), 1.43–1.37 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 162.9 (d, J = 245.4 Hz), 155.8, 140.9 (d, J = 6.2 Hz), 129.9 (d, J = 8.3 Hz), 125.2 (d, J = 2.0 Hz), 116.4 (d, J = 20.9 Hz), 113.3 (d, J = 21.0 Hz), 79.5, 62.5, 51.4, 41.5, 31.0, 29.1, 28.5 ppm; ^{19}F NMR (376 MHz, CDCl_3) δ –113.7 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{16}\text{H}_{25}\text{FNO}_3^+$ $[M + H]^+$ 298.1813, found 298.1814.

tert-Butyl (1-(2-Fluorophenyl)-5-hydroxypentan-2-yl)carbamate (3g): white solid (120 mg, 76%), mp 86–88 °C; IR (film) $\nu_{\max}$ 3340, 2979, 2933, 1689, 1491, 1456, 1363, 1231, 1170, 1054, 755 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.25–7.15 (m, 2H), 7.12–6.98 (m, 2H), 4.53–4.28 (m, 1H), 3.95–3.72 (m, 1H), 3.68–3.62 (m, 2H), 2.88–2.67 (m, 2H), 1.93 (s, 1H), 1.70–1.57 (m, 3H), 1.46–1.31 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 161.5 (d, J = 244.7 Hz), 155.8, 131.8, 128.2 (d, J = 8.1 Hz), 125.3 (d, J = 16.0 Hz), 124.1 (d, J = 3.2 Hz), 115.3 (d, J = 22.6 Hz), 79.3, 62.6, 51.1, 34.6, 31.3, 29.1, 28.4 ppm; ^{19}F NMR (376 MHz, CDCl_3 , rotamer) δ –117.6 (–117.8) ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{16}\text{H}_{25}\text{FNO}_3^+$ $[M + H]^+$ 298.1813, found 298.1819.

tert-Butyl (5-Hydroxy-1-(4-(trifluoromethyl)phenyl)pentan-2-yl)carbamate (3h): white solid (141 mg, 77%), mp 96–98 °C; IR (film) $\nu_{\max}$ 3360, 2931, 1683, 1524, 1328, 1246, 1157, 1116, 1064, 839 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.58–7.51 (m, 2H), 7.33–7.28 (m, 2H), 4.47–4.32 (m, 1H), 3.96–3.80 (m, 1H), 3.67–3.62 (m, 2H), 2.89–2.77 (m, 2H), 1.82 (s, 1H), 1.69–1.56 (m, 3H), 1.43–1.33 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.7, 142.6, 129.9, 128.9 (d, J = 32.4 Hz), 125.4, 125.3, 124.4 (d, J = 272.0 Hz), 79.6, 62.6, 51.4, 41.7, 31.2, 29.1, 28.4 ppm; ^{19}F NMR (376 MHz, CDCl_3) δ –62.4 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{17}\text{H}_{25}\text{F}_3\text{NO}_3^+$ $[M + H]^+$ 348.1781, found 348.1789.

tert-Butyl (5-Hydroxy-1-(4-(trifluoromethyl)phenyl)pentan-2-yl)carbamate (3i): white solid (148 mg, 80%), mp 58–60 °C; IR (film) $\nu_{\max}$ 3354, 2915, 2849, 1681, 1522, 1448, 1326, 1163, 1122, 1073, 701 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.52–7.35 (m, 4H), 4.47–4.32 (m, 1H), 3.95–3.79 (m, 1H), 3.67–3.62 (m, 2H), 2.91–2.75 (m, 2H), 1.79 (s, 1H), 1.70–1.57 (m, 3H), 1.43–1.32 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.7, 139.4, 132.9, 130.8 (d, J = 32.0 Hz), 128.9, 124.8 (d, J = 289.7 Hz), 124.3 (d, J = 272.2 Hz), 79.5, 62.6, 51.4, 41.6, 31.1, 29.1, 28.4 ppm; ^{19}F NMR (376 MHz, CDCl_3) δ –62.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{17}\text{H}_{25}\text{F}_3\text{NO}_3^+$ $[M + H]^+$ 348.1781, found 348.1785.

tert-Butyl (5-Hydroxy-1-(2-(trifluoromethyl)phenyl)pentan-2-yl)carbamate (3j): white solid (156 mg, 85%), mp 102–104 °C; IR (film) $\nu_{\max}$ 3356, 2983, 2933, 1683, 1526, 1312, 1167, 1116, 1054, 761 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.68–7.57 (m, 1H), 7.50–7.27 (m, 3H), 4.52–4.31 (m, 1H), 4.03–3.90 (m, 0.8H), 3.88–3.74 (m, 0.2H), 3.71–3.62 (m, 2H), 3.07–2.94 (m, 1H), 2.89–2.80 (m, 0.8H), 2.72–2.62 (m, 0.2H), 1.90 (s, 1H), 1.71–1.47 (m, 4H), 1.37–1.29 (m, 7.2H) 1.25–1.15 (m, 1.8H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3 , rotamers) δ 155.8, 137.5, 131.6 (d, J = 28.9 Hz) (132.4), 129.1 (129.4), 128.8 (128.5), 126.3 (d, J = 37.4 Hz), 124.7 (d, J = 266.1 Hz), 79.3, 62.6, 51.7 (53.1), 38.3 (40.0), 32.2, 29.1, 28.4 ppm; ^{19}F NMR (376 MHz, CDCl_3 , rotamer) δ –58.6 (–59.0) ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{17}\text{H}_{25}\text{NO}_3\text{F}_3^+$ $[M + H]^+$ 348.1781, found 348.1782.

tert-Butyl (1-(3-Chlorophenyl)-5-hydroxypentan-2-yl)carbamate (3k): white solid (145 mg, 87%), mp 99–101 °C; IR (film) $\nu_{\max}$ 3354, 2935, 1681, 1524, 1368, 1252, 1165, 1054, 790 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.26–7.15 (m, 3H), 7.13–7.01 (m, 1H), 4.48–4.31 (m, 1H), 3.90–3.77 (m, 1H), 3.67–3.61 (m, 2H), 2.80–2.69 (m, 2H), 1.88 (s, 1H), 1.69–1.55 (m, 3H), 1.45–1.36 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.8, 140.5, 134.2, 129.7, 129.7, 127.7, 126.7, 79.5, 62.5, 51.5, 41.4, 31.0, 29.1, 28.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{16}\text{H}_{25}\text{ClNO}_3^+$ $[M + H]^+$ 314.1518, found 314.1513.

tert-Butyl (1-(4-Cyanophenyl)-5-hydroxypentan-2-yl)carbamate (3l): white solid (132 mg, 83%), mp 108–110 °C; IR (film) $\nu_{\max}$ 3338, 2925, 2228, 1683, 1520, 1366, 1165 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.62–7.55 (m, 2H), 7.33–7.28 (m, 2H), 4.45–4.27 (m, 1H), 3.96–3.79 (m, 1H), 3.67–3.62 (m, 2H), 2.88–2.77 (m, 2H), 1.67–1.54 (m, 4H), 1.40–1.34 (m, 10H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.6, 144.2, 132.3, 130.3, 119.1, 110.5, 79.7, 62.5, 51.4, 42.2, 31.3, 29.1, 28.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{17}\text{H}_{25}\text{N}_2\text{O}_3^+$ $[M + H]^+$ 305.1860, found 305.1861.

Ethyl 4-(2-((tert-Butoxycarbonyl)amino)-5-hydroxypentyl)-benzoate (3m): white solid (158 mg, 85%), mp 52–54 °C; IR (film) $\nu_{\max}$ 3363, 2979, 2927, 1698, 1524, 1366, 1277, 1170, 1104, 1019, 771 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3) δ 8.01–7.93 (m, 2H), 7.28–7.23 (m, 2H), 4.48 (d, J = 8.4 Hz, 1H), 3.39–3.33 (m, 2H), 3.94–3.80 (m, 1H), 3.66–3.59 (m, 2H), 2.88–2.77 (m, 2H), 1.96 (s, 1H), 1.69–1.55 (m, 3H), 1.43–1.36 (m, 13H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 166.7, 155.7, 143.7, 129.7, 129.5, 128.8, 79.5, 62.5, 61.0, 51.4, 41.8, 31.0, 29.1, 28.5, 14.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{19}\text{H}_{30}\text{NO}_5^+$ $[M + H]^+$ 352.2118, found 352.2112.

tert-Butyl (5-Hydroxy-1-(naphthalen-1-yl)pentan-2-yl)carbamate (3n): yellow solid (60 mg, 34%), mp 75–77 °C; IR (film) $\nu_{\max}$ 3346, 2931, 1694, 1508, 1392, 1366, 1246, 1167, 1054, 784 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 8.22–8.05 (m, 1H), 7.89–7.69 (m, 2H), 7.57–7.45 (m, 2H), 7.41–7.25 (m, 2H), 4.60 (d, J = 7.9 Hz, 1H), 4.07–3.88 (m, 1H), 3.59–3.49 (m, 2H), 3.42–3.29 (m, 0.8H), 3.17–3.03 (m, 1.2H), 2.01 (s, 1H), 1.75–1.44 (m, 4H), 1.43–1.35 (m, 7.2H), 1.22–1.10 (m, 1.8H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.9, 134.6, 134.0, 132.4, 128.8, 127.7, 127.3, 126.2, 125.7, 125.4, 124.2, 79.3, 62.5, 51.2, 39.2, 31.0, 29.2, 28.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{20}\text{H}_{28}\text{NO}_3^+$ $[M + H]^+$ 330.2064, found 330.2066.

tert-Butyl (1-Hydroxytetradecan-4-yl)carbamate (3o): pale yellow oil (33 mg, 19%); IR (film) $\nu_{\max}$ 3330, 2927, 2853, 1694, 1528, 1366, 1250, 1176, 1052 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 5.58–5.48 (m, 0.3H), 5.18–5.08 (m, 0.3H), 4.46–4.24 (m, 1H), 3.97–3.75 (m, 1H), 3.69–3.63 (m, 2H), 3.62–3.39 (m, 1H), 2.22–2.11 (m, 0.7H), 1.97–1.89 (m, 0.7H), 1.64–1.54 (m, 3H), 1.46–1.43 (m, 10H), 1.32–1.23 (m, 16H), 0.94–0.83 (m, 3H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3 , rotamers) δ 156.0 (154.9), 79.0, 66.9, 62.7, 50.4, 35.7, 32.2, 31.9, 29.6, 29.3, 28.4, 28.3, 25.9, 22.7, 14.1 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{19}\text{H}_{40}\text{NO}_3^+$ $[M + H]^+$ 330.3003, found 330.3008.

tert-Butyl (1-Cyclohexyl-4-hydroxybutyl)carbamate (3p): pale yellow oil (35 mg, 24%); IR (film) $\nu_{\max}$ 3326, 2925, 1700, 1516, 1366, 1172, 1044 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 5.56–5.34 (m, 1H), 5.07 (d, J = 9.1 Hz, 1H), 4.04–3.93 (m, 1H), 3.72–3.64 (m, 2H), 1.96–1.86 (m, 2H), 1.78–1.67 (m, 5H), 1.61–1.51 (m, 2H), 1.50–1.44 (m, 9H), 1.36–1.23 (m, 6H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3 , rotamers) δ 155.0, 83.3 (82.6), 67.1, 31.9 (31.6), 30.2 (29.8), 28.5, 25.8, 24.8, 23.8 ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{15}\text{H}_{29}\text{NO}_3\text{Na}^+$ $[M + \text{Na}]^+$ 294.2040, found 294.2031.

tert-Butyl (6-Hydroxy-1-phenylhexan-2-yl)carbamate (4a): white solid (126 mg, 86%), mp 86–88 °C; IR (film) $\nu_{\max}$ 3327, 2931, 2857, 1681, 1450, 1363, 1172, 695 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3) δ 7.31–7.16 (m, 5H), 4.35 (d, J = 8.4 Hz, 1H), 1H), 3.89–3.77 (m, 1H), 3.64–3.57 (m, 2H), 2.83–2.70 (m, 2H), 1.59–1.44 (m, 5H), 1.43–1.33 (m, 11H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.8, 138.4, 129.6, 128.5, 126.4, 79.3, 62.8, 51.5, 41.6, 34.2, 32.5, 28.5, 22.3, ppm; HRMS (ESI-Orbitrap) m/z calcd for $\text{C}_{17}\text{H}_{28}\text{NO}_3^+$ $[M + H]^+$ 294.2064, found 294.2067.

tert-Butyl (6-Hydroxy-1-(o-tolyl)hexan-2-yl)carbamate (4b): white solid (130 mg, 84%), mp 65–67 °C; IR (film) $\nu_{\max}$ 3330, 2931, 2863, 1689, 1528, 1390, 1363, 1250, 1172, 1058, 740 cm^{-1} ; ^1H NMR (400 MHz, CDCl_3 , rotamers) δ 7.16–7.08 (m, 4H), 4.44–4.16 (m, 1H), 3.88–3.67 (m, 1H), 3.63–3.57 (m, 2H), 2.81–2.69 (m, 2H), 2.34 (s, 3H), 1.67 (s, 1H), 1.59–1.47 (m, 4H), 1.42–1.31 (m, 11H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3) δ 155.7, 136.8, 136.7, 130.4, 130.3, 126.5, 125.9, 79.2, 62.7, 50.9, 39.5, 34.6, 32.5, 28.5, 22.3,

19.7 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{18}H_{30}NO_3^+$ [$M + H$] $^+$ 308.2220, found 308.2224.

tert-Butyl (1-(2-Fluorophenyl)-6-hydroxyhexan-2-yl)carbamate (4c): white solid (117 mg, 75%), mp 84–86 °C; IR (film) $\nu_{\max}$ 3338, 2933, 2865, 1687, 1489, 1361, 1227, 1170, 753 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 7.25–7.12 (m, 2H), 7.11–6.95 (m, 2H), 4.45–4.15 (m, 1H), 3.88–3.69 (m, 1H), 3.64–3.58 (m, 2H), 2.86–2.60 (m, 2H), 1.71 (s, 1H), 1.62–1.47 (m, 4H), 1.42–1.31 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 161.5 (d, $J = 244.7$ Hz), 155.7, 131.8 (d, $J = 3.7$ Hz), 128.2 (d, $J = 8.2$ Hz), 125.4 (d, $J = 15.9$ Hz), 124.1 (d, $J = 3.3$ Hz), 115.3 (d, $J = 22.6$ Hz), 79.2, 62.8, 51.2, 34.6, 32.5, 28.5, 22.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, rotamer) δ –117.7 (–117.9) ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{17}H_{27}NO_3F^+$ [$M + H$] $^+$ 312.1970, found 312.1972.

tert-Butyl (6-Hydroxy-1-(2-(trifluoromethyl)phenyl)hexan-2-yl)carbamate (4d): White solid (93 mg, 51%), mp 72–74 °C; IR (film) $\nu_{\max}$ 3334, 2935, 1689, 1524, 1366, 1316, 1167, 1122, 1060, 767 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 7.68–7.57 (m, 1H), 7.49–7.28 (m, 3H), 4.47–4.31 (m, 0.8H), 4.29–4.16 (m, 0.2H), 4.00–3.86 (m, 0.8H), 3.85–3.73 (m, 0.2H), 3.67–3.59 (m, 2H), 3.06–2.94 (m, 1H), 2.88–2.79 (m, 0.8H), 2.69–2.58 (m, 0.2H), 1.70–1.41 (m, 7H), 1.37–1.29 (m, 7.2H) 1.24–1.17 (m, 1.8H) ppm; $^{13}C\{^1H\}$ NMR (150 MHz, $CDCl_3$, rotamers) δ 155.7, 137.7, 131.7 (d, $J = 26.3$ Hz) (132.4), 129.0 (d, $J = 29.4$ Hz) (129.0), 126.3 (d, $J = 53.8$ Hz), 124.8 (d, $J = 273.6$ Hz) (124.3), 79.2 (79.7), 62.8, 51.8 (53.2), 38.3 (39.9), 35.4 (35.9), 32.5 (31.6), 28.4, 22.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$, rotamer) δ –58.6 (–59.0) ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{18}H_{27}NO_3F_3^+$ [$M + H$] $^+$ 362.1938, found 362.1941.

tert-Butyl (6-Hydroxy-1-(*m*-tolyl)hexan-2-yl)carbamate (4e): white solid (86 mg, 55%), mp 78–80 °C; IR (film) $\nu_{\max}$ 3429, 2913, 2847, 1685, 1506, 1363, 1172, 759 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 7.22–7.14 (m, 1H), 7.07–6.93 (m, 3H), 4.42–4.14 (m, 1H), 3.89–3.75 (m, 1H), 3.64–3.57 (m, 2H), 2.80–2.64 (m, 2H), 2.35–2.29 (m, 3H), 1.71 (s, 1H), 1.60–1.46 (m, 4H), 1.44–1.32 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 155.8, 138.3, 138.0, 130.4, 128.3, 127.1, 126.6, 79.2, 62.7, 51.6, 41.5, 34.1, 32.5, 28.5, 22.3, 21.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{18}H_{30}NO_3^+$ [$M + H$] $^+$ 308.2220, found 308.2228.

tert-Butyl (1-(3-Fluorophenyl)-6-hydroxyhexan-2-yl)carbamate (4f): white solid (147 mg, 93%), mp 114–116 °C; IR (film) $\nu_{\max}$ 3356, 2927, 2847, 1681, 1524, 1366, 1248, 1170, 1050, 782, 695 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$) δ 7.27–7.21 (m, 1H), 7.00–6.86 (m, 3H), 4.35 (d, $J = 8.1$ Hz, 1H), 3.88–3.74 (m, 1H), 3.65–3.60 (m, 2H), 2.80–2.71 (m, 2H), 1.63–1.46 (m, 5H), 1.44–1.35 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 162.9 (d, $J = 245.4$ Hz), 155.7, 141.0 (d, $J = 7.1$ Hz), 129.9 (d, $J = 8.3$ Hz), 125.2 (d, $J = 2.1$ Hz), 116.4 (d, $J = 21.0$ Hz), 113.3 (d, $J = 21.0$ Hz), 79.4, 62.8, 51.5, 41.4, 34.3, 32.5, 28.5, 22.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$) δ –113.8 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{17}H_{27}FNO_3^+$ [$M + H$] $^+$ 312.1970, found 312.1973.

tert-Butyl (6-Hydroxy-1-(3-(trifluoromethyl)phenyl)hexan-2-yl)carbamate (4g): white solid (174 mg, 96%), mp 81–83 °C; IR (film) $\nu_{\max}$ 3334, 2929, 1687, 1528, 1328, 1167, 1120, 1075, 701 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 7.52–7.35 (m, 4H), 4.41–4.21 (m, 1H), 3.90–3.76 (m, 1H), 3.65–3.59 (m, 2H), 2.88–2.75 (m, 2H), 1.68 (s, 1H), 1.61–1.47 (m, 4H), 1.43–1.32 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, rotamers) δ 155.6, 139.4, 132.9, 130.7 (d, $J = 32.4$ Hz), 128.9 (128.4), 124.8 (d, $J = 294.5$ Hz) (125.7), 123.3 (123.0), 79.5, 62.7, 51.5, 41.4, 34.2, 32.4, 28.4, 22.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$) δ –62.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{18}H_{26}F_3NO_3Na^+$ [$M + Na$] $^+$ 384.1757, found 384.1760.

tert-Butyl (1-(3-Chlorophenyl)-6-hydroxyhexan-2-yl)carbamate (4h): white solid (151 mg, 92%), mp 109–111 °C; IR (film) $\nu_{\max}$ 3358, 2927, 2857, 1681, 1526, 1370, 1165, 1044, 784 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 7.26–7.15 (m, 3H), 7.10–7.02 (m, 1H), 4.41–4.15 (m, 1H), 3.87–3.72 (m, 1H), 3.66–3.60 (m, 2H), 2.78–2.68 (m, 2H), 1.64–1.46 (m, 5H), 1.45–1.35 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 155.7, 140.5, 134.2, 129.7,

127.7, 126.6, 79.4, 62.7, 51.5, 41.4, 34.2, 32.5, 28.5, 22.3 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{17}H_{27}ClNO_3^+$ [$M + H$] $^+$ 328.1674, found 328.1675.

tert-Butyl (6-Hydroxy-1-(*p*-tolyl)hexan-2-yl)carbamate (4i): white solid (142 mg, 91%), mp 80–82 °C; IR (film) $\nu_{\max}$ 3342, 2929, 2865, 1690, 1512, 1364, 1246, 1174, 1060 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$) δ 7.13–7.03 (m, 4H), 4.34 (d, $J = 8.1$ Hz, 1H), 3.86–3.73 (m, 1H), 3.63–3.57 (m, 2H), 2.78–2.64 (m, 2H), 2.32 (s, 3H), 1.63–1.45 (m, 5H), 1.44–1.33 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 155.8, 135.9, 135.2, 129.5, 129.1, 79.2, 62.8, 51.5, 41.1, 34.1, 32.5, 28.5, 22.3, 21.1 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{18}H_{30}NO_3^+$ [$M + H$] $^+$ 308.2220, found 308.2226.

tert-Butyl (1-(4-Fluorophenyl)-6-hydroxyhexan-2-yl)carbamate (4j): white solid (140 mg, 90%), mp 92–94 °C; IR (film) $\nu_{\max}$ 3336, 2929, 1687, 1510, 1363, 1221, 1174 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 7.17–7.10 (m, 2H), 7.02–6.93 (m, 2H), 4.39–4.16 (m, 1H), 3.85–3.72 (m, 1H), 3.64–3.59 (m, 2H), 2.77–2.68 (m, 2H), 1.70 (s, 1H), 1.58–1.45 (m, 4H), 1.44–1.35 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 161.7 (d, $J = 244.2$ Hz), 155.7, 134.1, 130.9 (d, $J = 7.9$ Hz), 115.2 (d, $J = 21.2$ Hz), 79.4, 62.7, 51.6, 40.9, 34.2, 32.5, 28.5, 22.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$) δ –117.1 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{17}H_{27}FNO_3^+$ [$M + H$] $^+$ 312.1970, found 312.1972.

tert-Butyl (6-Hydroxy-1-(4-(trifluoromethyl)phenyl)hexan-2-yl)carbamate (4k): white solid (179 mg, 99%), mp 95–97 °C; IR (film) $\nu_{\max}$ 3363, 2917, 2845, 1681, 1522, 1326, 1163, 1120, 1064, 757 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 7.58–7.51 (m, 2H), 7.33–7.27 (m, 2H), 4.41–4.17 (m, 1H), 3.91–3.78 (m, 1H), 3.66–3.60 (m, 2H), 2.86–2.77 (m, 2H), 1.62–1.47 (m, 5H), 1.42–1.33 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$, rotamers) δ 155.7, 142.7, 129.9, 128.8 (d, $J = 32.4$ Hz) (128.9), 125.4, 125.3, 124.4 (d, $J = 271.9$ Hz), 79.5, 62.7, 51.5, 41.6, 34.3, 32.5, 28.5, 22.3 ppm; ^{19}F NMR (376 MHz, $CDCl_3$) δ –62.4 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{18}H_{26}F_3NO_3Na^+$ [$M + Na$] $^+$ 384.1757, found 384.1754.

tert-Butyl (1-(4-cyanophenyl)-6-hydroxyhexan-2-yl)carbamate (4l): white solid (150 mg, 94%), mp 73–75 °C; IR (film) $\nu_{\max}$ 3344, 2933, 2226, 1687, 1524, 1366, 1250, 1172, 1054 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 7.62–7.55 (m, 2H), 7.33–7.28 (m, 2H), 4.41–4.22 (m, 1H), 3.91–3.74 (m, 1H), 3.65–3.60 (m, 2H), 2.86–2.77 (m, 2H), 1.66 (s, 1H), 1.59–1.47 (m, 4H), 1.40–1.34 (m, 11H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 155.6, 144.3, 132.3, 130.3, 119.0, 110.4, 79.5, 62.6, 51.5, 42.0, 34.5, 32.4, 28.4, 22.3 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{18}H_{27}N_2O_3^+$ [$M + H$] $^+$ 319.2016, found 319.2014.

Ethyl 4-(2-((tert-Butoxycarbonyl)amino)-6-hydroxyhexyl)-benzoate (4m): white solid (175 mg, 96%), mp 74–76 °C; IR (film) $\nu_{\max}$ 3363, 2933, 1710, 1687, 1522, 1363, 1273, 1172, 1103, 1021, 761 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$) δ 8.01–7.93 (m, 2H), 7.28–7.23 (m, 2H), 4.45–4.31 (m, 3H), 3.91–3.78 (m, 1H), 3.63–3.58 (m, 2H), 2.88–2.76 (m, 2H), 1.72 (s, 1H), 1.60–1.46 (m, 4H), 1.44–1.34 (m, 14H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 166.7, 155.7, 143.8, 129.7, 129.6, 128.8, 79.4, 62.7, 61.0, 51.5, 41.7, 34.2, 32.5, 28.5, 22.3, 14.5 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{20}H_{32}NO_5^+$ [$M + H$] $^+$ 366.2275, found 366.2279.

tert-Butyl (1-Hydroxypentadecan-5-yl)carbamate (4n): pale yellow oil (14 mg, 8%); IR (film) $\nu_{\max}$ 3340, 2925, 2851, 1691, 1524, 1363, 1246, 1176, 1052 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$) δ 4.35–4.21 (m, 1H), 3.67–3.62 (m, 2H), 3.59–3.50 (m, 1H), 1.79 (s, 1H), 1.65–1.52 (m, 3H), 1.48–1.41 (m, 14H), 1.30–1.24 (m, 16H), 0.90–0.85 (m, 3H) ppm; $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 156.0, 79.1, 62.9, 50.6, 35.6, 32.7, 32.0, 29.7, 29.5, 28.6, 26.0, 22.8, 22.1, 14.3 ppm; HRMS (ESI-Orbitrap) m/z calcd for $C_{20}H_{42}NO_3^+$ [$M + H$] $^+$ 344.3159, found 344.3158.

tert-Butyl (1-Cyclohexyl-5-hydroxypentyl)carbamate (4o): pale yellow oil (20 mg, 14%); IR (film) $\nu_{\max}$ 3332, 2936, 1706, 1524, 1364, 1159, 1038, 613 cm^{-1} ; 1H NMR (400 MHz, $CDCl_3$, rotamers) δ 5.47–5.31 (m, 1H), 5.10–4.88 (m, 1H), 4.04–3.90 (m, 1H), 3.71–3.61 (m, 2H), 1.97–1.86 (m, 2H), 1.75–1.66 (m, 3H), 1.63–1.52 (m, 4H), 1.50–1.42 (m, 11H), 1.40–1.18 (m, 6H) ppm; $^{13}C\{^1H\}$

NMR (100 MHz, CDCl₃) δ 155.3, 84.5, 81.3, 79.9, 62.7, 32.4, 32.3, 30.7, 30.6, 28.4, 25.9, 23.9, 23.9, 21.4 ppm; HRMS (ESI-Orbitrap) m/z calcd for C₁₆H₃₂NO₃⁺ [M + H]⁺ 286.2376, found 286.2376.

tert-Butyl (1-Phenyl-5-((trimethylsilyloxy)pentan-2-yl)-carbamate (5): colorless oil. ¹H NMR (400 MHz, CDCl₃) δ 7.30–7.26 (m, 2H), 7.22–7.16 (m, 3H), 4.54–4.43 (m, 1H), 3.85–3.74 (m, 1H), 3.57–3.53 (m, 2H), 2.85–2.77 (m, 1H), 2.76–2.71 (m, 1H), 1.68–1.48 (m, 4H), 1.40 (s, 9H), 0.09 (s, 9H) ppm.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.joc.9b01545.

Copies of ¹H NMR, ¹³C NMR, and ¹⁹NMR spectrum for all compounds (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank the National Natural Science Foundation of China (21772027 to B.-G.W. and 21702032 to C.-M.S.) for financial support. The authors also thank Dr. Han-Qing Dong (Arvinas, Inc.) for helpful suggestions.

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